

Nutrition Assistant Project

Phase 4: Project Planning

Introduction

Project Planning is an essential phase that defines the development roadmap for the Nutrition Assistant application. It includes selecting appropriate technologies, scheduling project activities, allocating responsibilities, identifying risks, and tracking milestones. Proper planning helps ensure the project is completed within the expected time, budget, and quality standards.

Project Objectives

Develop a secure and scalable nutrition tracking application. Deliver all project modules within the planned timeline. Ensure effective collaboration among team members. Monitor project progress using milestones and sprint reviews.

Technology Stack

Frontend: React.js, Vite, HTML5, CSS3, Tailwind CSS, JavaScript for creating a responsive and interactive user interface.

Backend: Node.js with Express.js to develop RESTful APIs and handle business logic.

Database: MongoDB for storing user accounts, meal logs, calorie information, and nutrition history.

Hosting: Vercel for frontend deployment and Render for backend hosting with cloud-based scalability.

Project Timeline

The project is divided into multiple development phases. The first week focuses on requirement analysis and UI design. The second week covers backend development and database integration. The third week is dedicated to frontend integration, API testing, and dashboard implementation. The final week focuses on testing, bug fixing, optimization, deployment, and documentation.

Milestones and Sprint Planning

Sprint 1: Repository setup, project structure, UI wireframes, and component design. **Sprint 2:** Backend API development, authentication module, and MongoDB integration. **Sprint 3:** Frontend integration, state management, food search, dashboard, and meal logging. **Sprint 4:** Testing, performance optimization, deployment, and final project documentation.

Risk Management

Potential risks include delays in API development, database integration issues, third-party API limitations, and deployment challenges. These risks are minimized through regular code reviews, sprint meetings, version control using Git, continuous testing, and backup plans for critical tasks.

Expected Deliverables

The planning phase produces a project schedule, sprint plan, milestone tracker, technology stack documentation, development roadmap, and resource allocation plan. These deliverables guide the implementation team throughout the project lifecycle.

Conclusion

The Project Planning phase establishes a structured workflow for the Nutrition Assistant project. By defining technologies, timelines, milestones, and risk mitigation strategies, the team can efficiently manage resources and successfully deliver a high-quality application.